Sahaj Bhadja

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Summary

Full-Stack Software Engineer pursuing an MSCS at USC with 2+ years building **AI applications**, **developer APIs**, and scalable products using **Python**, **TypeScript**, React, FastAPI, AWS, Docker, and CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Advanced NLP, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*

- Architected a multi-tenant **AI platform** using Azure SQL, Blob Storage, and row-level security for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through query optimization.
- Restructured fault-tolerant asynchronous LLM orchestration with **FastAPI** workers, processing 1K+ jobs through explicit state transitions and failure isolation while increasing throughput threefold under concurrent production workloads without cascading failures.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*

- Built a full-stack grading product with **React**, Flask, and multimodal AI, serving 150+ instructors and 10K+ daily requests while reducing feedback turnaround 60% through automated rubric workflows.
- Designed concurrent Python APIs with AWS RDS and secure file ingestion, processing 1K+ files per batch through fault-tolerant execution, validation, CI/CD releases, and stable customer-facing peak-load performance.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *LA, California*

- Built a Python pipeline processing 100K+ trajectory logs, improving long-horizon planning quality 18% through reproducible data preparation, model training, simulation evaluation, automated testing, and detailed experiment analysis.
- Architected a spatial-temporal encoder integrating map geometry and historical agent trajectories through multi-head attention, improving closed-loop planning stability 15% while validating model behavior across complex autonomous-driving scenarios.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Optimized large-scale API integrations using OOP patterns, AWS Glue, and S3, improving response times 30%; led distributed data workflows that increased pipeline efficiency 68% and reduced technical debt 20%.
- Engineered cross-platform microservices with automated builds and 20+ unit and integration tests using Jenkins, PyTest, and Mockito, supporting reliable releases across three operating systems within Agile sprints.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Developed a MERN platform with WebSockets, REST APIs, and eight microservices, synchronizing 2M+ records across authentication, billing, permissions, and notification workflows while reducing real-time data conflicts 33%.
- Automated **Docker** packaging and Jenkins CI/CD deployments, cutting release time 50%, while Azure SQL and analytics insights reduced resource waste 42% and increased customer lead conversion 15%.

Skills

Languages: Python, TypeScript, JavaScript, Java, SQL, C++, C#, JSON
Frontend: React, Next.js, HTML, CSS, TailwindCSS, Shadcn, Responsive Design, Component Development
Backend & AI: FastAPI, Flask, Node.js, REST APIs, SDK Development, LLMs, PyTorch, Transformers, FAISS
Data & Cloud: PostgreSQL, MySQL, SQL Server, MongoDB, Redis, AWS, Azure, GCP, Docker, Kubernetes, Distributed Systems
Engineering: Git, GitHub Actions, Jenkins, CI/CD, Unit Testing, Integration Testing, PyTest, Code Reviews, Debugging, Agile/Scrum

Projects

InferFlow SDK | TypeScript, AI Inference APIs, Node.js

- Built and tested a typed **TypeScript inference SDK** with asynchronous job submission, status polling, bounded exponential retries, timeout cancellation, stable error types, generic outputs, and mocked transport tests.

Big-Brain Live | GCP, PostgreSQL, FAISS, REST APIs

- Built a semantic document retrieval system over 50K+ documents using transformer embeddings and FAISS, enabling contextual search and summarization with sub-second retrieval through optimized indexing and similarity pipelines.

PiXMod Live | Next.js, TypeScript, MongoDB, Transformers

- Built an **AI-powered full-stack SaaS** with Next.js, TypeScript, MongoDB, CLIP, and Transformers, processing 100K+ image embeddings while supporting semantic search, subscription billing, and responsive customer-facing editing workflows.